Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026 **Phase 1:**

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1. Team & Project Overview

Team Name	Fresh team
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG4,SDG8,SDG10
Project / Solution Name	PivotPath AI
Name (Leader)	CHEN YANG
Email & Phone (Leader)	cylk080722@gmail.com
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

University graduates face growing uncertainty in the job market amid rapid AI disruption. Many students lack clear insights into future career trends and cannot assess how AI shocks at different intensity levels will reshape job demands. Without visualized forecasts, they struggle to select skills and career paths that align with sustainable development goals.

This issue mainly impacts undergraduate students, especially fresh graduates and career switchers. The mismatch between student skill preparation and evolving labour market needs raises unemployment risks and hinders inclusive, decent employment. It creates social and economic challenges: wasted educational resources, unequal career opportunities, and slow progress toward SDG 4 (quality education), SDG 8 (decent work and economic growth) and SDG 10 (reduced inequalities).

Traditional career guidance only offers static advice, ignoring the uncertain, multi-scenario nature of AIdriven labour market change. Our team addresses this gap to help young people, including disadvantaged learner groups, make informed, forward-looking career decisions and narrow employment inequality.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Our solution, PivotPath AI, is an AI-powered career forecasting dashboard for university students. It lets users assess employment risks under low, medium and high AI-impact scenarios and align career recommendations with SDG 4, SDG 8 and SDG 10.

We build the web app with React, TypeScript and Vite. The Express backend uses pdf-parse to extract text from uploaded CVs. We rely on CSV datasets and the O*NET 31.0 database for skill-job mapping, plus Remotive and World Bank APIs for labour market data. Browser LocalStorage saves user profiles and scenario preferences. Our custom i18n system supports Chinese, English and Malay trilingual switching, and we add rate, size and timeout limits to prevent API abuse.

From workshops, we learned GenAI workflow design, Replit cloud deployment, dataset integration and automated testing via Node Test Runner. We also practised custom internationalisation, backend API building and multi-scenario visualisation to link career paths to SDG targets. The platform turns uncertain labour market predictions into clear visuals, helping students make inclusive, sustainable career choices.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Our prototype frontend is built with React, TypeScript and Vite, with custom CSS styling, custom i18n for Chinese, English and Malay trilingual support, and front-end routing across seven pages. The Express backend provides APIs for PDF extraction via pdf-parse, with rate, file-size and timeout limits to prevent abuse.

We use CSV files and the O*NET 31.0 dataset for occupation-skill mapping. Remotive and World Bank public APIs supply labour market macro data. Browser LocalStorage stores user profiles and preferences.

We adopt Node Test Runner and end-to-end browser tests for validation. The whole system is developed and deployed on Replit, the cloud platform provided by SDC.

Our project does not use the SDC sponsor-provided GenAI API, but we adopt Replit, the cloud platform officially provided by SDC sponsors, for our development and deployment.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We followed design-thinking steps to build our prototype. First, we empathised with university students, identifying their anxiety about uncertain AI-driven job changes and the need for multilingual career guidance. We defined the core pain point: lack of visualised multi-scenario employment forecasts linked to SDGs.

We ideated and iterated the dashboard layout, reserving separate modules for CV uploading, scenario charts and SDG explanations. User experience was prioritised with our Chinese-English-Malay trilingual interface and simple form workflows.

We built and tested incrementally on Replit, writing unit and end-to-end tests to guarantee stability. The finished prototype is fully deployable, ready for further user testing and improvement in Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse percapita environmental impact of cities, focusing on air quality and waste management."*	Target 10.2: By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion or economic or other status. Target 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Over 16% of youth in Southeast Asia are NEET (not in employment, education or training). 22.9% of jobs are exposed to generative AI impacts, and 3.3% of jobs face high displacement risk. Meanwhile, about 35% of new graduates suffer from skill-job mismatch and lack AIaligned career guidance.

Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. Career matching success rate, in percentage (%) 2. Time for users to complete the career planning workflow, in minutes (min)
Metric Category	**Your Response**
	3. Coverage of occupation database, counted by number of occupations and industry categories
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Prototype demonstration phase target: A test set is constructed from 50 simulated fresh-graduate resumes and 20 job descriptions (JDs), with 3 reviewers manually annotating the AI job-matching results. The manual screening baseline Top-5 fit rate is 50%; after AI matching, the target is 60%, representing a 20% relative improvement in job-matching rate. At the user level, 10 fresh graduates are recruited for a real-name test to complete the task of "selecting 5 jobs that match their intentions." The traditional method takes an average of 50 minutes, while the target average time using this product is 30 minutes, a 40% reduction in screening time. The above are acceptance targets to be achieved during the prototype demonstration phase; actual testing will be completed before the demonstration and data will be updated according to actual results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs. [√]

The proposed solution targets a functional, deployed application rather than a theoretical business case study. [√]

All fields in this template have been completed within the word limits stated. [√]

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Replit	Cloud-based development platform and AI-assisted coding tool	Replit Terms of Service
O*NET® 31.0 Database	Public occupational and	Creative Commons Attribytion4.0

skills dataset International(CC BY4.0)

BLS Public Data API (v1) Public government dataset / API Public Domain (U.S. Government Work)

California Open Data Portal (OEWS CSV) Public government dataset Public Domain

Recharts Open-source code (Charting library) MIT License

Tailwind CSS Open-source code (CSS framework) MIT License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

陈昭 [Signature]

24/10/2026

[Date]